

DS 2500 *Fall* 2025 Lab Exercise 02

Deadline : 09/15/2025 11:59 PM EST

Late Deadline: 10/06/2025 11:59 PM EST with 50 points off

Total Points: 200 pts

Grading:

- This lab assignment is auto-graded. Gradescope can be a little picky, so make sure you don't put extraneous characters or whitespace in your answers -- and double-check the correct/incorrect confirmation.

Submission Instructions:

- Submit the Python file (`.py`) containing your solutions to Gradescope by the deadline specified at the top.
- You're allowed *unlimited* submissions.
- There are two deadlines. Submissions will not be accepted after the late deadline.

Disclosure: We've adapted lab problems from DS 2500 Spring 2025, so one or more problems may be the same or similar to those from that course.

Problem 1

1. Create a list `fruits` which contains 3 fruits in this order: "banana", "apple", and "orange".
2. Print the list.
3. Add "grape" to the list using the `.append()` method.

Problem 2

1. Write a function `clean_empty` that takes a list of strings and returns a list of integers. Convert each string to an integer, but if a string is empty (`""`), convert it to `0` instead. You can assume that each string is either a valid integer or an empty string.

Example:

```
clean_empty(["1",""]) #returns [1, 0]
clean_empty(["1","2","","3"]) #returns [1, 2, 0, 3]
```

Problem 3

1. Write a function `count_ints` that takes a 2D list of integers and a single integer and returns the number of occurrences of that integer in the list.

Example:

```
count_ints([[-1, -2, -3], [4, 5, 6]], 8) #returns 0
count_ints([[-1, -2, -3], [-1, 5, 6]], -1) #returns 2
count_ints([[-1, -2, -3, 4, 4], [4, 5, 6]], 4) #returns 3
```

Problem 4

1. Write a function `remove_duplicates` that takes a list and returns a new list after removing duplicates, *without using set()*.

Example:

```
remove_duplicates([1,1]) #returns [1]
remove_duplicates([1,2,3]) #returns [1, 2, 3]
remove_duplicates([3,1,2,3,2]) #returns [3, 1, 2]
```